

# Sakshi Sharma

7828386877 | [sakshi26sharma56@gmail.com](mailto:sakshi26sharma56@gmail.com) | [Linkedin](#) | [Github](#) | [leetcode](#) | [Portfolio](#)

Aspiring AI/ML engineer with hands-on experience in Machine Learning, Deep Learning, NLP, and Computer Vision, focused on building real-world AI solutions through innovation and practical implementation. Strong problem-solving mindset with experience working on research-driven, industry-oriented, and hackathon-based projects in emerging AI technologies..

## EDUCATION

B. Tech in Data Science (2023–2027) IPS Academy IES, Indore CGPA 9+  
12<sup>th</sup> Grade (2022) – Karnataka Vidya Niketan

## WORK EXPERIENCE

### THDCIL - Ministry of Power – September 2025 – Present

Working on backend and AI/ML pipelines for secure document verification. Implemented authenticity scoring, forgery detection, and hash-based tamper checks to automate verification workflows.

## PROJECTS

**Autonomous Vehicle for Calamity Rescue (AVCR)** – An ML-powered system deployed on Jetson Nano to assess survivability in disaster-hit, confined spaces. Uses real-time sensor data and models like Isolation Forest, LSTM, and Random Forest for detection, prediction, and classification.

### CarbonVault — AI Carbon Credit Platform

Full-stack sustainability platform for NGOs, admins, and corporates to automate carbon credit issuance. Integrated satellite imagery with NDVI analysis for automated land verification. Built with FastAPI, React, SpacetimeDB, and PostgreSQL.

### Auto Mail Apply

Agentic AI workflow that scrapes job listings, uses NLP to match candidate profiles, and auto-applies via email — end to end with zero manual effort. Built with FastAPI, Selenium, and Python NLP pipelines.

**Rockfall Prediction & Alert System (Open-Pit Mines)** – Developed an ML-based system to predict rockfall risks in open-pit mines using geological and vibration sensor data. Implemented risk-level classification and real-time alerting to enhance worker safety and prevent accidents.

**AI-Powered Personal Legal Assistant** – Developed a comprehensive web-based AI assistant with document summarization, legal drafting, case strategy advisory, and document classifier to streamline legal workflows. .

**Pothole Detection** – Built a computer vision solution using YOLO and OpenCV to detect potholes from street-level images. Aims to support smart infrastructure management and automated road safety monitoring.

## ACHIEVEMENTS

**Winner – Techno Maths Presentation Winner** – Science Exhibition 2024

**Smart India Hackathon (SIH) 2025** Finalist

**Udaan'26 – Winner**

**Winner – VyaparX Idea Pitching Competition**

**Top 20** – Clear Skies Innovation Challenge

**NEC (National Entrepreneurship Challenge) Finalist** **Top 100** Campus

Ambassador, IIT Bombay .

**Winner** – Inter-College Chess Competition 2023, 25

**Participations hackathons** – Bhartiya Antariksha Hackathon, Hacksungum, GDSC Codespire, Pravah 3.0, Hackdata'

25, Build with India, Samsung Solve for Tomorrow, Flipkart Grid 6.0 (Robotics), Snow Hacks

## SKILLS

### Programming & Scripting

Python, C, C++

### Machine Learning & AI

Machine Learning, Deep Learning (MLP, CNN, RNN), NLP, LSTM, LLM, Transformers, Hugging Face, LangChain, RAG, Prompt Engineering, Fine-tuning (LoRA, PEFT), Reinforcement Learning, Scikit-learn

### Computer Vision

OpenCV, YOLOv8, Image Segmentation, MediaPipe

### Data & Analytics

SQL, MS Excel, Power BI, Big Data, Data Analysis, EDA, MySQL, PostgreSQL, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

### Web & Backend

Django, FastAPI, HTML, CSS, Bootstrap, REST APIs, WebSockets

### Cloud & MLOps

Vercel, Docker, Google Colab, MLflow, AWS (familiar)

### Tools & Platforms

Git, GitHub, VS Code, Jetson Nano

### Databases

MySQL, PostgreSQL, ChromaDB (Vector DB)

### Soft Skills

Problem-solving, Leadership, Calm under pressure, Presentations, Agentic Workflow Design

## CERTIFICATION

Java Tutorial (IIT Bombay)

RDBMS PostgreSQL

Python Tutorial (IIT Bombay)

Artificial Intelligent and machine learning (NPTEL)

## LEADERSHIP

**Founder** – E-Cell, IES IPS Academy

**Core Member** – Bad Talk